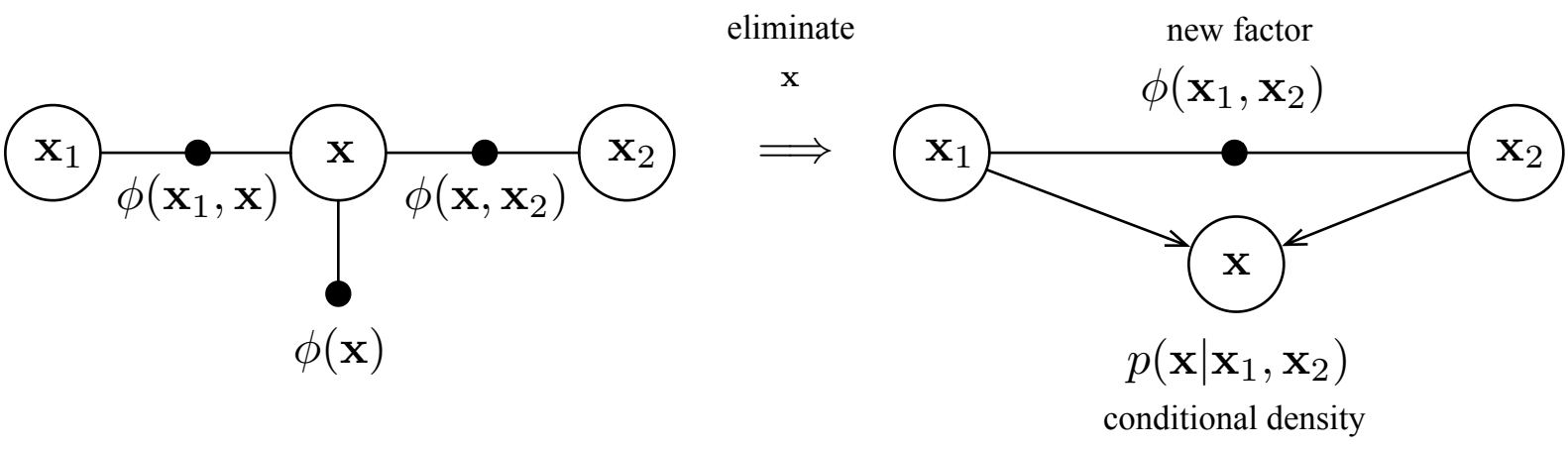
$\mathbf{x}_1$
$\mathbf{x}$
$\mathbf{x}_2$
$\phi(\mathbf{x}_1, \mathbf{x})$
$\phi(\mathbf{x}, \mathbf{x}_2)$
$\phi(\mathbf{x})$
eliminate
$\mathbf{x}$
$\Longrightarrow$
new factor
$\phi(\mathbf{x}_1, \mathbf{x}_2)$
$\mathbf{x}_1$
$\mathbf{x}_2$
$\mathbf{x}$
$p(\mathbf{x}|\mathbf{x}_1, \mathbf{x}_2)$
conditional density